The Born Rule from Recognition Science: $P = |\psi|^2$ from J-Cost Phase Invariance

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Abstract

We derive the Born rule $P(i) = |\psi_i|^2$ from Recognition Science (RS) by two independent routes. First, the RS path-weight derivation: the path action $C[\gamma] = \int J(r(t))\,dt$ yields weights $w = e^{-C}$; the amplitude–cost bridge $C = 2A$ gives $|\psi|^2 = e^{-C}$, recovering the Born rule directly from the cost functional. Second, a uniqueness theorem: the probability function must satisfy five structural requirements—(i) non-negativity, (ii) normalization, (iii) J-cost phase independence, (iv) multiplicativity from path-action additivity, and (v) continuity. We prove that the unique function satisfying all five is $P(c_i) = |c_i|^2$. Alternative rules $P = |c|^k$ with $k \neq 2$ are excluded. The interference identity follows automatically. The Born rule is not postulated but is a theorem of the cost-geometric framework. All structural results are derived from the RS axioms [1].

1 Introduction

The Born rule $P(i) = |\psi_i|^2$ is the link between the quantum formalism and experimental outcomes [2]. In standard quantum mechanics it is postulated. Why the probability is $|\psi|^2$ rather than $|\psi|$, $|\psi|^4$, or some other function of the amplitude is one of the deepest questions in quantum foundations [3].

Gleason's theorem (1957) derives $P = |\psi|^2$ from the properties of frame functions on Hilbert spaces of dimension ≥ 3 [4]. In Many-Worlds, Zurek's "envariance" argument derives the Born rule from entanglement symmetry [5]. RS [1] provides two independent routes to $P = |\psi|^2$:

(a) A *path-weight derivation*: the path action $C[\gamma] = \int J(r(t))\,dt$ yields weights $w = e^{-C}$; the amplitude–cost bridge $C = 2A$ gives $|\psi|^2 = e^{-C}$, recovering the Born rule.

(b) A *uniqueness derivation*: non-negativity, normalization, phase independence, multiplicativity over independent events, and continuity jointly force $P = |c|^2$ as the unique probability function.

We present both in this paper. Route (a) is the RS-native derivation; route (b) is a structural uniqueness theorem that serves as an independent cross-check.

2 Cost Functional Foundation

RS derives all physical law from a single cost functional [1]. The three axioms are:

(A1) $J(1) = 0$ *(Normalization)*

(A2) $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$ *(Recognition Composition Law)*

(A3) $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$ *(Calibration)*

Theorem 2.1 (Cost Uniqueness [1]). *The unique continuous solution to (A1)–(A3) is $J(x) = \frac{1}{2}(x + x^{-1}) - 1$, satisfying $J(x) \geq 0$ with equality iff $x = 1$.*

Proof sketch. Setting $H(t) = 1 + J(e^t)$ reduces (A2) to the d'Alembert equation $H(s + t) + H(s - t) = 2H(s)H(t)$. With $H(0) = 1$ and $H''(0) = 1$, the Aczél classification forces $H(t) = \cosh t$, yielding $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

The path-weight derivation connects the cost functional to quantum amplitudes via Feynman's path integral [6], replacing the classical action with the RS cost J.

3 Path-Weight Derivation

The RS-native derivation proceeds from the path action.

Definition 3.1 (Path Action). For a recognition path γ on the ledger, the path action is:

$$C[\gamma] = \int_\gamma J(r(t))\, dt, \tag{1}$$

where $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ is the cost functional and $r(t)$ is the scale ratio along the path.

Definition 3.2 (Path Weight and Amplitude). Each path carries a weight $w(\gamma) = e^{-C[\gamma]}$ and an amplitude:

$$\psi(\gamma) = e^{-C[\gamma]/2} \cdot e^{i\phi(\gamma)}, \tag{2}$$

where $\phi(\gamma)$ is the phase accumulated on the 8-tick cycle. The modulus–cost bridge is $C = 2A$ where $A = -\ln|\psi|$, so $|\psi|^2 = e^{-C} = w$.

Proposition 3.3 (Born Rule from Path Weights — Structural Derivation). *The probability of outcome i is:*

$$\boxed{P(i) = |\psi_i|^2 = \frac{e^{-C_i}}{\sum_j e^{-C_j}} = \frac{w_i}{\sum_j w_j},} \tag{3}$$

where $\psi_i = \sum_{\gamma \to i} \psi(\gamma)$ is the total amplitude for paths reaching outcome i.

Proof. The cost of reaching outcome i via a set of paths is C_i, the total path action. The weight $w_i = e^{-C_i}$ is positive and well-defined. Normalization requires $\sum_i P(i) = 1$, so $P(i) = w_i / \sum_j w_j$. Since $w_i = |\psi_i|^2$ by the modulus–cost bridge, this is $P(i) = |\psi_i|^2 / \sum_j |\psi_j|^2$. For normalized states ($\sum_j |\psi_j|^2 = 1$), this reduces to $P(i) = |\psi_i|^2$. □

Remark 3.4 (Status of the path-weight derivation). The path-weight derivation above is structurally motivated but partly circular: Definition 3.2 *defines* $\psi(\gamma) = e^{-C[\gamma]/2} \cdot e^{i\phi}$, so $|\psi|^2 = e^{-C}$ by construction. The modulus-cost bridge $C = 2A$ (with $A = -\ln|\psi|$) is therefore definitional, not a derived identity. The non-circular content of this derivation is that the path-action principle selects e^{-C} as the natural weight for cost-minimizing dynamics; the identification of this weight with the quantum probability is the RS structural postulate. The uniqueness derivation in Section 5 (Theorem 5.1) provides a fully independent, non-circular derivation of $P = |c|^2$.

Remark 3.5. The phase $\phi(\gamma)$ arises from the 8-tick cycle: position on the $2^D = 8$ clock determines the phase of each path contribution. The path weight $w = e^{-C}$ depends only on the cost (magnitude), not on the phase. This is the origin of J-cost phase invariance—phase carries temporal position information but does not affect the probability of an outcome.

4 The Five Requirements

As an independent cross-check, we derive the Born rule from five structural requirements on any probability function.

Definition 4.1 (Probability Function)**.** A probability function P assigns to each complex amplitude c_i a real number $P(c_i)$ representing the probability of outcome i.

Proposition 4.2 (Structural Requirements)**.** *The probability function must satisfy:*

(i) ***Non-negativity****:* $P(c_i) \geq 0$ *for all* c_i*.*

(ii) ***Normalization****:* $\sum_i P(c_i) = 1$ *whenever* $\sum_i |c_i|^2 = 1$*.*

(iii) ***Phase independence****:* $P(c_i) = P(|c_i|)$*.*

(iv) ***Multiplicativity****: For independent recognition events with amplitudes* c_1 *and* c_2*, the joint probability factors:* $P(c_1 c_2) = P(c_1)\, P(c_2)$*.*

(v) ***Continuity****:* P *is a continuous function of the amplitude modulus.*

Justification. Requirements (i) and (ii) are standard. Requirement (iii) follows from the J-cost functional: the cost $J(r)$ is defined on positive reals (scale ratios), so the cost assigned to an amplitude $c = re^{i\theta}$ is $J(r) = J(|c|)$. Since cost determines probability through the path-weight bridge, and cost is phase-blind, probability is phase-blind.

Requirement (iv) follows from the additivity of path actions: for independent events on disjoint ledger regions, the total path action is $C_{12} = C_1 + C_2$. Therefore $w_{12} = e^{-C_{12}} = e^{-C_1}\, e^{-C_2} = w_1\, w_2$. Since $P \propto w = |\psi|^2$, probabilities of independent events multiply. At the amplitude level, this means $P(c_1 c_2) = P(c_1)\, P(c_2)$.

Requirement (v) is physically motivated: measurable quantities (here, the probability of an outcome as a function of the amplitude) vary continuously under continuous changes of experimental parameters. Mathematically, it is needed to solve the Cauchy functional equation uniquely (without continuity, pathological solutions exist that require the Axiom of Choice). □

5 Uniqueness of $P = |c|^2$

Theorem 5.1 (Born Rule from Uniqueness)**.** *The unique function satisfying requirements (i)–(v) is:*

$$\boxed{P(c_i) = |c_i|^2.} \tag{4}$$

Proof. By phase independence (iii), $P(c) = f(|c|)$ for some function $f : [0, \infty) \to [0, \infty)$.

Step 1: f is a power law. By multiplicativity (iv), for any $r_1, r_2 > 0$:

$$f(r_1 r_2) = f(r_1)\, f(r_2). \tag{5}$$

This is Cauchy's multiplicative functional equation on $(0, \infty)$ [8]. By requirement (v) (continuity), with f non-negative and $f(r) > 0$ for $r > 0$ (required for non-trivial probabilities), the unique solution is $f(r) = r^k$ for some constant $k > 0$.

Step 2: $k = 2$ from normalization. Consider a state $|\psi\rangle = \alpha|a\rangle + \beta|b\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. Normalization (ii) requires $f(|\alpha|) + f(|\beta|) = 1$. Setting $|\alpha|^2 = p$ and $|\beta|^2 = 1 - p$:

$$p^{k/2} + (1 - p)^{k/2} = 1 \quad \text{for all } p \in [0, 1]. \tag{6}$$

At $p = 1/2$: $2 \cdot (1/2)^{k/2} = 1$, so $(1/2)^{k/2} = 1/2$, giving $k/2 = 1$ and hence $\boldsymbol{k = 2}$.

Verification: for $k = 2$, $p^{k/2} + (1 - p)^{k/2} = p + (1 - p) = 1$ for all $p \in [0, 1]$, confirming that $k = 2$ satisfies normalization identically. ✓

Note: The argument uses the normalization convention $|\alpha|^2 + |\beta|^2 = 1$. This convention is itself tied to the Born rule: we are proving that the probability function *consistent* with this normalization is $P = |c|^2$. If one normalizes amplitudes differently (e.g., $|\alpha| + |\beta| = 1$), the analysis changes. The RS derivation fixes the normalization convention through the path-weight bridge: $\sum_i e^{-C_i} = 1$ in the vacuum, which is the cost-functional analog of $\sum_i |c_i|^2 = 1$.

Step 3: Exclusion of alternatives. For $k = 1$: $\sqrt{p} + \sqrt{1 - p} \neq 1$ in general (at $p = 1/2$: $\sqrt{2} \neq 1$). For $k = 4$: $p^2 + (1 - p)^2 \neq 1$ in general (at $p = 1/2$: $1/2 \neq 1$). Only $k = 2$ satisfies normalization for all p. □

Remark 5.2. Requirements (i)–(iii) alone do not force $P = |c|^2$. The functional equation $g(p) + g(1 - p) = 1$ (where $g(p) = f(\sqrt{p})$) has many continuous solutions beyond $g(p) = p$. It is requirement (iv)—multiplicativity from the additivity of path actions—that restricts f to power laws (given continuity from (v)), and then normalization selects $k = 2$. This is why the J-cost structure is essential: without the cost functional forcing additivity, the Born rule would not be uniquely determined.

6 Interference

Lemma 6.1 (Interference Identity)**.** *For amplitudes $c_1, c_2 \in \mathbb{C}$:*

$$|c_1 + c_2|^2 = |c_1|^2 + |c_2|^2 + 2\,\mathrm{Re}(c_1^* c_2). \tag{7}$$

The cross-term $2\,\mathrm{Re}(c_1^* c_2)$ is the interference term.

Proof. $|c_1 + c_2|^2 = (c_1 + c_2)(c_1 + c_2)^* = |c_1|^2 + c_1 c_2^* + c_2 c_1^* + |c_2|^2 = |c_1|^2 + |c_2|^2 + 2\,\mathrm{Re}(c_1^* c_2)$. □

Corollary 6.2 (Constructive and Destructive Interference)**.** *If c_1 and c_2 are* in phase *($\arg c_1 = \arg c_2$), then $|c_1 + c_2|^2 = (|c_1| + |c_2|)^2$ (constructive interference). If they are* out of phase *($\arg c_1 = \arg c_2 + \pi$), then $|c_1 + c_2|^2 = (|c_1| - |c_2|)^2$ (destructive interference).*

7 Why Not $|\psi|$ or $|\psi|^4$?

The exclusion of alternative probability rules deserves emphasis, as it is a common question.

Proposition 7.1 ($|\psi|$ Is Excluded)**.** *The rule $P(c) = |c|$ violates normalization.*

Proof. For an equal superposition $|\psi\rangle = \frac{1}{\sqrt{2}}(|a\rangle + |b\rangle)$, $P(a) + P(b) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2} \neq 1$. □

Proposition 7.2 ($|\psi|^4$ Is Excluded)**.** *The rule $P(c) = |c|^4$ violates normalization.*

Proof. For the same state, $P(a) + P(b) = (1/\sqrt{2})^4 + (1/\sqrt{2})^4 = 1/4 + 1/4 = 1/2 \neq 1$. □

Corollary 7.3. *Any rule $P = |c|^k$ with $k \neq 2$ violates normalization for at least one state. The Born rule is the unique survivor.*

8 Comparison with Other Derivations

	Gleason	**Zurek**	**RS**
Starting point	Frame functions on Hilbert space	Envariance (entanglement symmetry)	J-cost path weights
Dimension	≥ 3	Any	Any
Interpretation	Any	Many-Worlds	Ledger Realism
Assumes	Hilbert space structure	Entanglement axioms	Cost functional + 8-tick phase
Multiplicativity	Implicit	Derived	From path-action additivity
Result	$P = \|\psi\|^2$	$P = \|\psi\|^2$	$P = \|\psi\|^2$

Remark 8.1. Gleason's theorem requires the Hilbert space dimension ≥ 3; it does not apply to qubits (dimension 2). The RS uniqueness theorem (Theorem 5.1) applies to *any* dimension, including qubits, because it relies on cost-functional multiplicativity rather than properties of projection-valued measures on a Hilbert lattice. Hardy's [7] derivation from five reasonable axioms and the RS derivation are structurally complementary: Hardy's most restrictive axiom ("simplicity") corresponds to the RS statement that J is uniquely determined by (A1)–(A3). Deutsch [9] and Wallace derive the Born rule from decision-theoretic axioms within the Everett framework; the RS derivation replaces the decision-theoretic apparatus with the J-cost multiplicativity, which has a more direct physical motivation (additivity of path actions). All approaches arrive at $P = |\psi|^2$.

9 Predictions

1. **Born rule is exact.** No deviations from $P = |\psi|^2$ at any energy scale, including the Planck scale. The rule follows from the structure of the cost functional, which does not change with energy.

2. **No alternative probability rules.** $P = |\psi|$, $|\psi|^4$, and all other power laws are excluded (Section 7). Any experiment showing $P \neq |\psi|^2$ would falsify the J-cost framework.

3. **Interference is structural.** All quantum interference phenomena follow from the cross-term $2\,\mathrm{Re}(c_1^* c_2)$ in the interference identity (Section 6), which is a consequence of the complex amplitude structure forced by the 8-tick cycle.

4. **No hidden-variable corrections.** The Born rule is not an approximation or an ensemble average—it is the exact probability of individual outcomes. RS predicts no sub-quantum deviations of the kind sought by nonlinear quantum mechanics proposals.

10 Conclusion

The Born rule is derived from RS by two independent routes. The path-weight derivation (Proposition 3.3) proceeds directly from the cost functional: path actions are additive, weights are multiplicative, and the amplitude–cost bridge $C = 2A$ gives $P = |\psi|^2$. The uniqueness derivation (Theorem 5.1) shows that non-negativity, normalization, phase independence, multiplicativity, and continuity jointly force $P = |c|^2$ as the unique probability function. The interference identity follows automatically from the complex amplitude structure. The Born rule is not postulated but is a theorem of the cost-geometric framework.

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